SDG BLOCKCHAIN ACCELERATOR

Impact Measurement Framework

SolarVillage: A Hydra-based Energy Settlement & Carbon Issuance Architecture

1. Project Information

- **Project Name:** SolarVillage
- **Challenge & UNDP Office:** UNDP Nigeria
- **Document Version:** v1.0 (Dec 23, 2025)

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2. SDG Alignment

This project is explicitly designed to deliver **measurable, pilot-stage contributions** to priority Sustainable Development Goals, with a focus on outcomes that can be verified through system-generated data rather than self-reported claims. SolarVillage aligns its technical architecture, operational processes, and reporting framework to SDGs where **direct attribution is possible during a controlled pilot**, ensuring that impact claims are proportionate, defensible, and suitable for UNDP monitoring and learning.

SolarVillage contributes directly to these three UN Sustainable Development Goals:

SDG 7 – Affordable and Clean Energy

Target / Indicator:

- **Indicator 7.1.1:** Proportion of population with access to electricity

Contribution: SolarVillage increases access to affordable and reliable electricity through solar mini-grids, supported by verifiable delivery records. Metered energy consumption is tracked at 15-minute intervals, enabling transparent reporting on household-level electricity access and usage.

SDG 9 – Industry, Innovation and Infrastructure

Target / Indicator:

- **Indicator 9.1.1:** Proportion of rural population with access to basic infrastructure

Contribution: The project deploys digital public infrastructure for operating, monitoring, and financing rural energy systems. This includes real-time settlement, transparent revenue tracking, and auditable system logs, strengthening the operational and financial sustainability of mini-grids.

SDG 13 – Climate Action

Target / Indicator:

- **Indicator 13.2.2:** Total greenhouse gas emissions per year

Contribution: By replacing diesel and kerosene-based energy generation with solar mini-grids, SolarVillage enables measurable emissions displacement. Energy delivery data is structured and retained in a form suitable for future carbon credit verification, aligned with international and national carbon standards.

3. Key Performance Indicators (KPIs)

The Key Performance Indicators defined below translate high-level development objectives into **operational metrics that can be observed, measured, and audited during the pilot period**. KPIs are deliberately limited to indicators that rely on verifiable system data or structured feedback, enabling continuous monitoring, early risk detection, and evidence-based assessment of pilot success and scalability readiness.

The following KPIs are designed to capture **service delivery, system reliability, financial transparency, and carbon-readiness**, using objectively measurable pilot data.

KPI	Measurement Method	Target / Threshold	Notes
Households Served	Registered meters with active usage	≥ pilot baseline (site-specific)	Disaggregated by village
Electricity Delivered (kWh)	Meter data aggregated per grid	Baseline + monthly growth	Recorded in 15-minute intervals
System Uptime	Meter availability logs	≥ 95% uptime	Excluding force majeure
Successful Payment Rate	Confirmed payments vs attempts	≥ 90%	Across all payment channels
Activation Latency	Time from payment → meter activation	< 5 minutes (online)	Offline codes may vary
Operator Revenue Visibility	On-chain + dashboard completeness	100% of pilot transactions	Audit-ready
Carbon Readiness Coverage	% of energy data meeting MRV standards	≥ 90%	Measures readiness, not issuance
User Satisfaction	Structured feedback sessions	> 70% positive	Qualitative + simple scoring
Reporting Completeness	Dashboards populated as designed	100%	UNDP-facing views

4. Tracking Methods

SolarVillage employs a **mixed on-chain and off-chain tracking** approach to ensure that all KPIs can be monitored consistently and cross-validated. This approach prioritizes transparency, data integrity, and auditability, while remaining practical for deployment in rural and low-connectivity environments. Tracking methods are designed to support both real-time operational oversight and periodic UNDP reporting.

The following impact data is collected using a combination of **on-chain system records**, **off-chain operational logs**, and **structured feedback mechanisms**.

On-chain Metrics

- Settlement transaction success rate
- Energy accounting records (Wh aggregation)
- Revenue visibility and completeness
- Period-level aggregation outputs supporting carbon readiness

Off-chain Metrics

- Metering system APIs (usage, uptime)
- Backend dashboards (payments, activation times)
- Structured user feedback sessions
- Operator validation of meter data

All critical operational and financial metrics are cross-checked between on-chain logs and dashboards on a regular basis to ensure consistency.

5. Baseline vs Projected Outcomes

Given the pilot nature of the intervention, formal baselines are established at project start and used to evaluate progress over time rather than to demonstrate long-term impact prematurely. Projected outcomes focus on **directional improvement, system reliability, and readiness for scale**, providing UNDP and partners with a clear basis for assessing whether the solution merits further deployment or integration.

Metric	Baseline	Projected Outcome (Pilot Phase)
Carbon-ready energy data	0	≥ 90% of delivered energy
Households served	Site-specific	≥ baseline, growing monthly
Electricity delivered	0	Continuous increase
System uptime	N/A	≥ 95%

Metric	Baseline	Projected Outcome (Pilot Phase)
Successful payment rate	N/A	≥ 90%
Operator revenue visibility	0	100% coverage

6. Monitoring & Reporting Plan

The monitoring and reporting plan is structured to support **continuous learning, accountability, and timely course correction** throughout the pilot. Responsibilities, reporting frequency, and formats are clearly defined to ensure that performance data is accessible to implementers, operators, and UNDP Nigeria, while also generating auditable records suitable for partner review and future scale-up decisions.

The following monitoring and reporting approach is designed to support **continuous oversight, learning, and auditability**, while remaining practical for a pilot context.

Frequency

- **Operational monitoring:** Continuous
- **Internal reviews:** Monthly
- **Pilot reporting:** Quarterly

Responsible Parties

- **Solution Maker:** System monitoring, data aggregation, dashboards
- **Mini-grid Operators:** Meter data validation and operational inputs
- **UNDP Nigeria:** Oversight, review, learning documentation

Format

- Live dashboards (operational + impact views)
- Monthly internal summary notes
- Quarterly pilot progress reports

Tools

- SolarVillage dashboards
- Cardano / Hydra transaction logs
- Metering system APIs

- Structured feedback notes and session summaries

7. Risks & Mitigation

This section identifies key risks that could affect pilot delivery, adoption, or impact measurement, and pairs each risk with **practical mitigation measures appropriate to a controlled pilot environment**. The risk framework reflects both technical and non-technical considerations and emphasizes early detection, redundancy, and stakeholder coordination to reduce execution risk and protect development outcomes.

The following risks reflect **pilot-relevant technical, operational, and coordination challenges**, with corresponding mitigation strategies.

Risk	Mitigation
Low user adoption	Community onboarding, simplified user flows, multiple payment methods
Technical integration delays	Modular rollout, staged integrations, fallback options
Meter data inconsistencies	Validation rules, operator review, redundancy
Payment failures	Multiple payment rails, retry logic, offline options
Regulatory uncertainty	Early engagement with REA and NCCC; audit-ready system design
Carbon methodology delays	Parallel alignment with multiple standards; readiness-first approach
Partner coordination challenges	Clear roles, regular check-ins, UNDP convening
Over-reliance on donor funding	Emphasis on operator viability and future revenue streams